SUPPLY CHAIN ANALYTICS

End-to-End Power BI Intelligence Report

Sales & Revenue | Procurement & Supplier | Inventory Analytics | Logistics & Fulfillment

Period: 2020 – 2024 | Regions: Asia Pacific, Europe, Latin America, North America

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1. Executive Summary

This report delivers a comprehensive analysis of a multi-dimensional Supply Chain dataset built in Microsoft Power BI. The solution spans five years (2020–2024) across four global regions and covers the full supply chain lifecycle — procurement, inventory, fulfillment, and sales — across four dedicated dashboards with advanced interactivity including Drill Down, Drill Through, and custom Tooltips.

Metric	Value	Dashboard
Total Revenue	₹ 442.09 Million	Sales & Revenue
Total Profit	₹ 220.97 Million	Sales & Revenue
Total Orders	250K | Avg Order Value: ₹8.88K	Sales & Revenue
Total Purchase Cost	₹ 6.3 Billion	Procurement & Supplier
On-Time Delivery %	22.52% (Critical Risk)	Procurement & Supplier
Total Inventory Value	₹ 208.7 Billion	Inventory Analytics
Units on Hand	815 Million	Inventory Analytics
Total Quantity Shipped	4.97 Million | Fill Rate: 79.38%	Logistics & Fulfillment
Shipment Orders	50K | Avg Shipment Time: 10.51 days	Logistics & Fulfillment

2. Dataset Overview & Data Model

The dataset follows a Star Schema with 3 Fact Tables and 6 Dimension Tables, enabling fast DAX calculations and seamless cross-filtering across all report pages.

2.1 Fact & Dimension Tables

Table	Type	Key Fields
Fact_Sales	Fact	Sale ID, Customer ID, Product ID, Geography ID, Date
Fact_Procurement	Fact	PO Number, Product ID, Supplier ID, Warehouse ID, Order Date
Fact_Inventory_Snapshots	Fact	Product ID, Warehouse ID, Date
Fact_Order_Fulfillment	Fact	Order ID, Customer ID, Warehouse ID, Carrier, Shipment Date
Dim_Date	Dimension	Date, Date Key, Month, Month Number, Quarter, Year
Dim_Customer	Dimension	Customer ID, Customer Name, Customer Tier, Geography ID
Dim_Product	Dimension	Product ID, Brand, Category, SKU ID, Sub Category, Reorder Point
Dim_Geography	Dimension	Geography ID, City, Country, Region, State/Province
Dim_Warehouse	Dimension	Warehouse ID, Region, Country, Max Capacity, Is Cross Docking

Table	Type	Key Fields
Dim_Supplier	Dimension	Supplier ID, Supplier Name, Lead Time Mean Days, Sustainability Rating

2.2 Key Relationships

Dim_Date acts as the central time spine connected to all 4 fact tables. Dim_Geography links Fact_Sales and Dim_Customer. Dim_Product connects Fact_Sales, Fact_Procurement, and Fact_Inventory_Snapshots. Dim_Warehouse bridges Fact_Inventory_Snapshots and Fact_Order_Fulfillment. All relationships are One-to-Many with single-direction filter flow, ensuring model integrity and optimal query performance.

2.3 DAX Measure Categories

- Time Intelligence: Revenue LY, Profit LY, Revenue YoY%, Revenue MoM%, Revenue Last Month, Profit YoY%
- Sales KPIs: Total Revenue, Total Profit, Total Orders, Total Discount, Avg Order Value, Avg Profit per Order
- Procurement KPIs: Avg Delay Time, Avg Delivery Time, On-Time Delivery %, Total Purchase Cost, Total Freight Cost, Total Purchase Orders
- Inventory KPIs: Units on Hand, Units in Transit, Safety Stock Level, Capacity Utilization %, Total Inventory Value
- Logistics KPIs: Fill Rate %, Avg Shipment Time, Total Quantity Shipped, Shipment Orders, Total Quantity Ordered

3. Sales & Revenue Performance Dashboard

3.1 KPI Cards

KPI	Value	Insight
Profit Performance	₹220.97M (Goal: ₹220.91M)	Marginally above target — healthy margin management
Revenue Performance	₹442.09M (Goal: ₹441.13M)	Tracking slightly above target
Total Orders	250K	High transaction volume across all regions
Avg Order Value	₹8.88K	Driven by Gold and Silver tier customers

3.2 Charts & Insights

- Scatter Chart — Total Discount & Total Profit by City: Strong positive correlation between discounts and profit per city. Cities offering up to ₹200K in discounts generate profits near ₹15M. Drill Down applied: Region → Country → City to identify that a few APAC cities drive disproportionate profit. Drill Through configured from city data points to a city-level transaction detail page.
- Bar Chart — Total Revenue by Brand: DeltaWorks leads at ₹403M, BetaCo at ₹378M. ThetaPro lags at ₹155M — flagged for brand strategy review. Tooltips show units sold, avg discount %, and profit margin on hover.
- Line Chart — Total Profit by Year: Profit peaked at ~₹224M in 2021 before dipping, recovering through 2024. Drill Down (Year → Quarter → Month) identifies which quarters drove the decline — correlated with 2021–22 supply chain disruptions.
- Bar Chart — Total Revenue by Region: Asia Pacific dominates at ₹831M, North America ₹585M, Europe ₹535M, Latin America ₹267M. Cross-filtered by Region slicer across all pages.
- Donut Chart — Revenue by Customer Tier: Bronze 49.0% (₹1.08Bn), Silver 35.3% (₹782.8M), Gold 15.7% (₹349.1M). Bronze dominance signals an upsell opportunity. Drill Through from tier segment navigates to a full customer list with order history.

4. Procurement & Supplier Performance Dashboard

4.1 KPI Cards

KPI	Value	Insight
Avg Delay Time	5.48 days	Moderate — requires monitoring to prevent stockouts
Avg Delivery Time	37.51 days	Long lead times, critical for safety stock planning
On-Time Delivery %	22.52%	⚠ Critical — only 1 in 4 orders arrive on time
Total Purchase Orders	10K	Substantial procurement across 15 suppliers
Avg Quality Inspection	5.48 / 10	Below acceptable threshold — supplier audits required
Total Purchase Cost	₹6.3 Billion	Total spend across all POs over 5 years

4.2 Charts & Insights

- Line Chart — Total Freight Cost by Year: Rose from ₹5.08M (2020) to ₹5.2M (2022 peak), dropped sharply to ₹4.9M (2023), partially recovering in 2024. Drill Down (Year → Quarter) identifies cost-spike quarters for supplier renegotiation.
- Bar Chart — Total Orders by Supplier ID: SUP013 leads with 706 orders, SUP008 follows at 703. Distribution is even (623–706), indicating no single-supplier dependency risk. Drill Through from a supplier bar navigates to a PO-level detail page with quality scores and delay analysis.
- Area Chart — Total Purchase Cost (Monthly, 2020–2024): Monthly costs fluctuate between ₹50M and ₹100M+, with spikes in mid-2021 and late-2024. Drill Down (Year → Month) pinpoints cost anomalies. Tooltips display exact monthly cost, freight cost, and on-time delivery % on hover.

5. Inventory Analytics Dashboard

5.1 KPI Cards

KPI	Value	Insight
Capacity Utilization %	1.14K	Near-full or over-capacity warehouses — expansion needed
Total Inventory Value	₹208.7 Billion	Massive working capital locked in stock
Safety Stock Level	56 Million	Adequate buffer to absorb supply disruptions
Warehouse Capacity	714K units	Total theoretical capacity across 5 warehouses
Units on Hand	815 Million	Current physical stock — exceeds stated capacity
Units in Transit	25 Million	Goods in movement across the supply network

5.2 Charts & Insights

- Horizontal Bar Chart — Units in Transit by Warehouse: WH003 and WH004 each hold 5.0M units in transit; WH005 slightly lower at 4.9M. Near-equal distribution reflects balanced planning. Tooltips show transit age, expected arrival, and associated freight cost per warehouse.
- Line Chart — Units on Hand by Month: Consistent upward trend from ~60M (January) to ~82M (December) with a February dip. Suggests progressive inventory build-up — possible year-end procurement front-loading. Drill Down (Year → Month) investigates which product categories drive the build-up.
- Clustered Bar Chart — Units on Hand by Warehouse: WH003 leads at 167M, WH002 trails at 159M. Near-equal distribution signals effective network balancing. Drill Through from a warehouse bar navigates to a product-level stock breakdown with capacity heat map.

6. Logistics & Order Fulfillment Dashboard

6.1 KPI Cards

KPI	Value	Insight
Shipment Orders	50K	Total shipment transactions in the period
Total Quantity Shipped	4.97 Million	Cumulative units dispatched to customers
Fill Rate %	79.38%	⚠ Below 80% — ~1 in 5 orders partially fulfilled
Avg Shipment Time	10.51 days	Dispatch-to-receipt time across all carriers

6.2 Charts & Insights

- Clustered Bar Chart — Shipped & Ordered Qty by Year: Ordered quantities (1.2M–1.3M) consistently exceed shipped quantities (1.0M) each year. Chronic gap of ~300K units signals a structural fulfillment capacity constraint. Drill Down (Year → Quarter → Month) pinpoints when gaps are most severe.
- Donut Chart — Shipped Qty by Carrier: All 8 carriers hold near-equal share — FedEx 12.68% (630K), down to Maersk 12.27% (610K). Deliberate multi-carrier diversification eliminates single-carrier risk. Tooltips show avg shipment time and fill rate per carrier on hover.
- Line Chart — Shipped Qty by Year: Wave pattern — peak ~1,020K (2021), dip in 2022, recovery in 2023, another drop in 2024. Drill Down (Year → Quarter) identifies demand or capacity shocks per quarter.

- Horizontal Bar Chart — Shipped Qty by Warehouse: WH001–WH003 and WH005 each at 1.00M; WH004 marginally lower at 0.98M. Reflects strong warehouse capacity parity across the network.
- Stacked Bar Chart — Shipped Qty by Method: Road 2.49M (50%), Sea 1.49M (30%), Air 0.99M (20%). Road dominance reflects regional delivery patterns. Drill Down (Method → Carrier) applied to understand carrier-mode composition. Tooltips show avg cost per unit and avg delivery days by method.

7. Interactive Features: Drill Down, Drill Through & Tooltips

7.1 Drill Down — Hierarchy Paths

Dashboard	Chart	Drill Down Path
Sales & Revenue	Scatter — Discount & Profit by City	Region → Country → City
Sales & Revenue	Line — Total Profit by Year	Year → Quarter → Month
Procurement	Line — Total Freight Cost	Year → Quarter → Month
Procurement	Area — Total Purchase Cost	Year → Month
Inventory	Line — Units on Hand by Month	Year → Month → Week
Inventory	Bar — Units on Hand by Warehouse	Warehouse → Product Category → SKU
Logistics	Bar — Shipped & Ordered by Year	Year → Quarter → Month
Logistics	Line — Shipped by Year	Year → Quarter
Logistics	Stacked Bar — Method	Method → Carrier

7.2 Drill Through — Navigation Paths

Source Visual	Drill Through Target	Filter Applied
Scatter — City Data Point	City Detail Page	City Name
Donut — Customer Tier	Customer Tier Detail Page	Customer Tier
Bar — Revenue by Brand	Brand Performance Page	Brand Name
Bar — Orders by Supplier	Supplier Detail Page	Supplier ID
Bar — Units by Warehouse	Warehouse Detail Page	Warehouse ID
Donut — Carrier Share	Carrier Detail Page	Carrier Name

7.3 Custom Tooltips

Chart	Tooltip Content
Bar — Revenue by Brand	Units Sold, Avg Discount %, Profit Margin %, YoY Growth
Donut — Revenue by Customer Tier	Avg Order Value, Total Orders, Avg Discount per Tier
Area — Purchase Cost Monthly	Monthly Freight Cost, On-Time Delivery %, Avg Quality Score
Donut — Carrier Shipment Share	Avg Shipment Time (days), Fill Rate %, Total Units
Stacked Bar — Shipping Method	Avg Cost per Unit, Avg Delivery Days, % of Total Volume
Bar — Units in Transit by Warehouse	Transit Age, Expected Arrival Date, Freight Cost

8. Key Insights & Recommendations

Sales & Revenue

- Revenue and profit have both narrowly beaten targets — set more ambitious 2025 goals to sustain growth momentum. [KPI Cards, Sales Dashboard]
- Bronze customers drive 49% of revenue; a structured Bronze → Silver upsell program could significantly raise average order value. [Donut Chart — Revenue by Customer Tier]
- Asia Pacific at ₹831M is 42% larger than the next region — double down on APAC infrastructure and distribution. [Bar Chart — Revenue by Region]
- ThetaPro brand at ₹155M significantly underperforms peers — evaluate brand investment or discontinuation. [Bar Chart — Revenue by Brand]

Procurement & Supplier

- ⚠ On-Time Delivery of 22.52% is critically low. Renegotiate supplier SLAs immediately and introduce delivery penalty clauses. [KPI Card, Procurement Dashboard]
- Average quality inspection score of 5.48/10 is below threshold — audit top-volume suppliers SUP013 and SUP008 first. [Bar Chart — Orders by Supplier ID]
- Monthly purchase cost swings (₹50M–₹100M+) indicate reactive buying — shift to planned, volume-committed procurement to reduce cost volatility. [Area Chart — Purchase Cost Monthly]

Inventory

- Total Inventory Value of ₹208.7Bn ties up enormous working capital. A 5% reduction via demand forecasting improvement could free ₹10.4Bn. [KPI Card, Inventory Dashboard]
- Progressive Jan–Dec inventory build-up suggests front-loaded procurement cycles — review safety stock policy given the low on-time delivery rate. [Line Chart — Units on Hand by Month]

Logistics & Fulfillment

- ⚠ Fill Rate of 79.38% is below acceptable thresholds. Target 95%+ within 12 months through capacity expansion or 3PL partnerships. [KPI Card, Logistics Dashboard]
- The chronic 300K-unit annual fulfillment gap is structural, not seasonal — capital investment in warehouse capacity or 3PL overflow contracts is recommended. [Clustered Bar Chart — Shipped vs Ordered by Year]
- Air freight at 20% of volume should be audited — air is expensive and should be reserved for high-urgency, high-value shipments only. [Stacked Bar Chart — Shipped by Method]

9. Conclusion

This Power BI Supply Chain Analytics report translates five years of multi-dimensional operational data into actionable intelligence across Sales, Procurement, Inventory, and Logistics. The star schema data model, combined with Drill Down hierarchies (9 visuals), Drill Through navigation (6 paths), and contextual Tooltips (6 custom pages), provides every level of the organization — from C-suite to operations analysts — with self-service analytical capability.

The two most critical findings are the 22.52% On-Time Delivery rate and the 79.38% Fill Rate, both indicating systemic supply chain execution gaps that require immediate corrective action. Alongside these, the ₹208.7Bn inventory value presents the single largest working capital optimization opportunity in the dataset.

Priority	Action	Source
1 — Critical	Renegotiate supplier SLAs to address 22.52% On-Time Delivery	Procurement Dashboard — KPI Card
2 — Critical	Target Fill Rate 95%+ via capacity investment or 3PL partnerships	Logistics Dashboard — KPI Card & Bar Chart
3 — High	Launch Bronze → Silver customer upsell program	Sales Dashboard — Donut Chart
4 — High	Audit quality for SUP013, SUP008 (avg score 5.48/10)	Procurement Dashboard — Bar Chart
5 — Medium	Optimize inventory to free working capital from ₹208.7Bn pool	Inventory Dashboard — KPI Card

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